

Defect Classification — Model Comparison Report

Binary classification of Clean (0) vs Defective (1) samples · Original vs Expanded feature sets · Training rows: 800 | Testing rows: 200

Model Comparison Summary — Original Features

Aggregate performance metrics (for the Defective / positive class) across all trained models using the original feature set.

Model	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.724	0.420	0.532	0.666
Decision Tree	0.591	0.520	0.553	0.570
Random Forest	0.565	0.520	0.542	0.598
SVM	0.667	0.460	0.544	0.636
K-Nearest Neighbors	0.559	0.570	0.564	0.564
Gaussian Naive Bayes	0.821	0.230	0.359	0.644
Gradient Boosting	0.618	0.550	0.582	0.652
Extra Trees	0.564	0.530	0.546	0.565

Model Comparison Summary — Expanded Features

Aggregate performance metrics (for the Defective / positive class) across all trained models using the expanded feature set.

Model	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.586	0.650	0.616	0.632
Decision Tree	0.578	0.590	0.584	0.580
Random Forest	0.618	0.550	0.582	0.660
SVM	0.559	0.710	0.626	0.622
K-Nearest Neighbors	0.595	0.660	0.626	0.643
Gaussian Naive Bayes	0.765	0.260	0.388	0.640
Gradient Boosting	0.660	0.640	0.650	0.690
Extra Trees	0.633	0.570	0.600	0.646

Highlighted cells indicate the best-performing model for that metric within each feature set. The expanded feature set lifts Gradient Boosting to the top overall performer (F1 0.650, AUC-ROC 0.690), and generally improves recall on the Defective class across most models, though Gaussian Naive Bayes and several others still show weaker recall.

Per-Model Classification Reports

Original vs Expanded feature-set results shown side by side for each model.

Logistic Regression

Original Features					Expanded Features				
Class	Prec.	Recall	F1	Supp.	Class	Prec.	Recall	F1	Supp.
Clean (0)	0.59	0.84	0.69	100	Clean (0)	0.61	0.54	0.57	100
Defective (1)	0.72	0.42	0.53	100	Defective (1)	0.59	0.65	0.62	100
Accuracy			0.63	200	Accuracy			0.59	200
Macro Avg	0.66	0.63	0.61	200	Macro Avg	0.60	0.59	0.59	200
Weighted Avg	0.66	0.63	0.61	200	Weighted Avg	0.60	0.59	0.59	200

Decision Tree

Original Features					Expanded Features				
Class	Prec.	Recall	F1	Supp.	Class	Prec.	Recall	F1	Supp.
Clean (0)	0.57	0.64	0.60	100	Clean (0)	0.58	0.57	0.58	100
Defective (1)	0.59	0.52	0.55	100	Defective (1)	0.58	0.59	0.58	100
Accuracy			0.58	200	Accuracy			0.58	200
Macro Avg	0.58	0.58	0.58	200	Macro Avg	0.58	0.58	0.58	200
Weighted Avg	0.58	0.58	0.58	200	Weighted Avg	0.58	0.58	0.58	200

Random Forest

Original Features					Expanded Features				
Class	Prec.	Recall	F1	Supp.	Class	Prec.	Recall	F1	Supp.
Clean (0)	0.56	0.60	0.58	100	Clean (0)	0.59	0.66	0.63	100
Defective (1)	0.57	0.52	0.54	100	Defective (1)	0.62	0.55	0.58	100
Accuracy			0.56	200	Accuracy			0.60	200
Macro Avg	0.56	0.56	0.56	200	Macro Avg	0.61	0.60	0.60	200
Weighted Avg	0.56	0.56	0.56	200	Weighted Avg	0.61	0.60	0.60	200

SVM

Original Features

Expanded Features

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.59	0.77	0.67	100
Defective (1)	0.67	0.46	0.54	100
Accuracy			0.61	200
Macro Avg	0.63	0.61	0.61	200
Weighted Avg	0.63	0.61	0.61	200

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.60	0.44	0.51	100
Defective (1)	0.56	0.71	0.63	100
Accuracy			0.57	200
Macro Avg	0.58	0.57	0.57	200
Weighted Avg	0.58	0.57	0.57	200

K-Nearest Neighbors

Original Features

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.56	0.55	0.56	100
Defective (1)	0.56	0.57	0.56	100
Accuracy			0.56	200
Macro Avg	0.56	0.56	0.56	200
Weighted Avg	0.56	0.56	0.56	200

Expanded Features

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.62	0.55	0.58	100
Defective (1)	0.59	0.66	0.63	100
Accuracy			0.60	200
Macro Avg	0.61	0.60	0.60	200
Weighted Avg	0.61	0.60	0.60	200

Gaussian Naive Bayes

Original Features

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.55	0.95	0.70	100
Defective (1)	0.82	0.23	0.36	100
Accuracy			0.59	200
Macro Avg	0.69	0.59	0.53	200
Weighted Avg	0.69	0.59	0.53	200

Expanded Features

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.55	0.92	0.69	100
Defective (1)	0.76	0.26	0.39	100
Accuracy			0.59	200
Macro Avg	0.66	0.59	0.54	200
Weighted Avg	0.66	0.59	0.54	200

Gradient Boosting

Original Features

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.59	0.66	0.63	100
Defective (1)	0.62	0.55	0.58	100
Accuracy			0.60	200
Macro Avg	0.61	0.60	0.60	200
Weighted Avg	0.61	0.60	0.60	200

Expanded Features

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.65	0.67	0.66	100
Defective (1)	0.66	0.64	0.65	100
Accuracy			0.66	200
Macro Avg	0.66	0.66	0.65	200
Weighted Avg	0.66	0.66	0.65	200

Extra Trees

Original Features

Expanded Features

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.56	0.59	0.57	100
Defective (1)	0.56	0.53	0.55	100
Accuracy			0.56	200
Macro Avg	0.56	0.56	0.56	200
Weighted Avg	0.56	0.56	0.56	200

Class	Prec.	Recall	F1	Supp.
Clean (0)	0.61	0.67	0.64	100
Defective (1)	0.63	0.57	0.60	100
Accuracy			0.62	200
Macro Avg	0.62	0.62	0.62	200
Weighted Avg	0.62	0.62	0.62	200